



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Conclusion:

The Java program was successfully implemented to demonstrate the use of the final keyword with variables, methods, and classes. The final keyword was used to prevent modification of important data, ensuring data integrity and program reliability. The Bank Account and Library Book Management examples showed how final is useful for defining constant values such as account numbers and ISBNs that should not be changed after initialization. This experiment improved the understanding of data security and object-oriented programming concepts in Java.

Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

Code:

```
1 class BankAccount {  
2     final int accountNumber;  
3     String accountHolder;  
4     double balance;  
5  
6     BankAccount(int accountNumber, String accountHolder, double balance) {  
7         this.accountNumber = accountNumber;  
8         this.accountHolder = accountHolder;  
9         this.balance = balance;  
10    }  
11  
12    void display() {  
13        System.out.println("Account Number: " + accountNumber);  
14        System.out.println("Account Holder: " + accountHolder);  
15        System.out.println("Balance: " + balance);  
16    }  
17  
18    public static void main(String[] args) {  
19        BankAccount b = new BankAccount(123456, "Anurag Thakur", 50000);  
20  
21        b.display();  
22    }  
23 }
```

Output:

```
Account Number: 123456  
Account Holder: Anurag Thakur  
Balance: 50000.0  
  
=== Code Execution Successful ===
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Code:

```

1 class Book {
2     final String isbn;
3     String title;
4     String author;
5     double price;
6
7     Book(String isbn, String title, String author, double price) {
8         this.isbn = isbn;
9         this.title = title;
10        this.author = author;
11        this.price = price;
12    }
13
14    void display() {
15        System.out.println("ISBN: " + isbn);
16        System.out.println("Title: " + title);
17        System.out.println("Author: " + author);
18        System.out.println("Price: " + price);
19    }
20
21    public static void main(String[] args) {
22        Book b = new Book("978-9356061234", "Java Programming", "Anurag Thakur", 799);
23
24        b.display();
25    }
26 }

```

Output:

```

ISBN: 978-9356061234
Title: Java Programming
Author: Anurag Thakur
Price: 799.0

=== Code Execution Successful ===

```